

Midterm - Representations of Quivers

April 13th, 2026

Choose up to five problems from this list to submit.

1. Let Q be the quiver $1 \longleftarrow 2 \longrightarrow 3$ and let M be a representation

$$k \xleftarrow{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}} k^3 \xrightarrow{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}} k^2 .$$

Write M as a direct sum of simple representations, indecomposable projective and injective representations.

2. Let Q be the quiver

$$1 \begin{matrix} \xleftarrow{\alpha} \\ \xrightarrow{\beta} \end{matrix} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$1 \begin{matrix} \xleftarrow{1} \\ \xrightarrow{\lambda} \end{matrix} 2 \quad \lambda \in k;$$

$$1 \begin{matrix} \xleftarrow{1} \\ \xrightarrow{0} \end{matrix} 2 \quad \lambda = \infty.$$

- (a) Show that each M_λ is indecomposable.
 - (b) Show that $M_\lambda \simeq M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k .
 - (c) Show that $\text{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.
3. Let Q be a quiver, $L \xrightarrow{f} M \xrightarrow{g} N$ be an exact sequence in $\text{rep } Q$ and X be representation of Q . Show that $\text{Ker } g_* \subseteq \text{Im } f_*$, where f_* and g_* are defined in

$$\text{Hom}(X, L) \xrightarrow{f_*} \text{Hom}(X, M) \xrightarrow{g_*} \text{Hom}(X, N) .$$

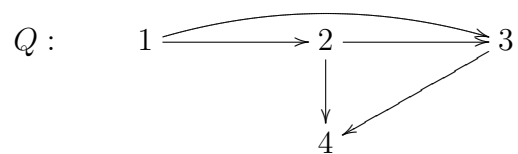
4. Let $g : L \rightarrow M$ be an injective morphism between representations of Q , and let $I(i)$ be the injective representation at vertex i . Show that the map

$$g^* : \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i))$$

is surjective.

5. Prove that any projective representation corresponding to vertex i , $P(i)$ is indecomposable.

6. Compute a projective resolutions for representations $S(1)$ and $S(2)$ for quiver



7. Let $M = (M_i, \psi_\alpha)$ be a representation of Q . Prove that for any vertex i in Q , there is an isomorphism of vector spaces:

$$\text{Hom}(M, I(i)) \simeq M_i.$$

8. Let i and j be vertices in the quiver Q . Prove that the vector space $\text{Hom}(I(i), I(j))$ has a basis consisting of all paths from j to i in Q . In particular, $\text{End}(I(i)) \simeq k$.